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Course: Calculus Beta

Assignment: MyLab Math - Uge 9

A probability density function of a random variable is given by $f(x) = 3x^{-4}$ on the interval $[1, \infty)$. Find the expected value, the variance, and the standard deviation.

First, find the expected value. If X is a continuous random variable with probability density function f on $[a, b]$, then the expected value of X is

$\mu = \int_a^b x f(x) dx$. Multiply the probability function by x and integrate over the domain of the function.

$$\mu = \int_1^{\infty} (3x^{-3}) dx$$

Since this is an improper integral, we will integrate from 1 to b and then find the limit as $b \rightarrow \infty$. Factor 3 out of the integral, and then find the antiderivative of x^{-3} .

$$\lim_{b \rightarrow \infty} \int_1^b (3x^{-3}) dx = 3 \lim_{b \rightarrow \infty} \left(-\frac{x^{-2}}{2} \right) \Big|_1^b$$

Now factor $-\frac{1}{2}$ out of the limit, substitute the limits of integration, and simplify.

$$\begin{aligned} 3 \lim_{b \rightarrow \infty} \left(-\frac{x^{-2}}{2} \right) \Big|_1^b &= -\frac{3}{2} \lim_{b \rightarrow \infty} (b^{-2} - 1^{-2}) \\ &= -\frac{3}{2} (0 - 1) \\ &= 1.5 \end{aligned}$$

The expected value is $\mu = 1.5$. Next, find the variance. If X is a continuous random variable with probability density function f on $[a, b]$, then

the variance of X is $\text{Var}(X) = \int_a^b (x - \mu)^2 f(x) dx$. It can also be found using the alternative formula $\text{Var}(x) = \int_a^b x^2 f(x) dx - \mu^2$.

We will use the alternative formula. Again, since this is an improper integral, we will integrate from 1 to b and then find the limit as $b \rightarrow \infty$.

$$\text{Var}(X) = \lim_{b \rightarrow \infty} \int_1^b (3x^{-2}) dx - 1.5^2$$

Factor 3 out of the integral, and then find the antiderivative of x^{-2} .

$$\lim_{b \rightarrow \infty} \int_1^b (3x^{-2}) dx - 1.5^2 = 3 \lim_{b \rightarrow \infty} (-x^{-1}) \Big|_1^b - 1.5^2$$

Now factor -1 out of the limit, substitute the limits of integration, and simplify.

$$\begin{aligned} 3 \lim_{b \rightarrow \infty} (-x^{-1}) \Big|_1^b - 1.5^2 &= -3 \lim_{b \rightarrow \infty} (b^{-1} - 1^{-1}) - 1.5^2 \\ &= -3 (0 - 1) - 1.5^2 \\ &= 0.75 \end{aligned}$$

Therefore, the variance is $\text{Var}(X) = 0.75$.

Finally, find the standard deviation. The standard deviation of X is $\sigma = \sqrt{\text{Var}(X)}$.

$$\sigma = \sqrt{0.75} \approx 0.87$$